

UNIT 2 - CORRELATION AND REGRESSION

Correlation:

The term correlation is used by a common man without knowing that he is making use of the term correlation. For example when parents advice their children to work hard so that they may get good marks, they are correlating good marks with hard work.

If two series vary in such a way, that fluctuations in one are accompanied by the fluctuations in the other, these variables are said to be correlated. Like rise in price of a commodity, reduces its demand and vica-versa.

The study related to the characteristics of only variable such as height, weight, ages, marks, wages, etc., is known as uni variate analysis. The statistical Analysis related to the study of the relationship between two variables is known as Bi-Variate Analysis. Sometimes the variables may be inter-related. In health sciences we study the relationship between blood pressure and age, consumption level of some nutrient and weight gain, total income and medical expenditure, etc., The nature and strength of relationship may be examined by correlation and Regression analysis.

Thus Correlation refers to the relationship of two variables or more. (e-g) relation between height of father and son, yield and rainfall, wage and price index, share and debentures etc.

Correlation is statistical Analysis which measures and analyses the degree or extent to which the two variables fluctuate with reference to each other. The word relationship is important. It indicates that there is some connection between the variables. It measures the closeness of the relationship. Correlation does not indicate cause and effect relationship. Price and supply, income and expenditure are correlated.

Correlation expresses the inter-dependence of two sets of variables upon each other. One variable may be called as (subject) independent and the other relative variable (dependent). Relative variable is measured in terms of subject.

Uses of correlation:

It is used in physical and social sciences.

It is useful for economists to study the relationship between variables like price, quantity etc. Businessmen estimates costs, sales, price etc. using correlation.

It is helpful in measuring the degree of relationship between the variables like income and expenditure, price and supply, supply and demand etc.

Sampling error can be calculated.

It is the basis for the concept of regression.

Types of Correlation : Correlation can be of following types -

- (i) **Positive and Negative Correlation :** If changes in two connected series is in the same direction, i.e. increase in one variable is associated with increase in other variable, the correlation is said to be positive. For example increase in father's age, increase in son's age.

If the two related series change in opposite direction i.e. increase in one variable is associated with the corresponding decrease in other variable, the correlation is said to be negative.

- (ii) **Linear and Non-Linear :** If the amount of change in one variable tends to bear constant ratio of change in the other variable, the correlation is said to be linear. We get a straight line if the variables of these series are marked on graph paper.

Correlation would be called non-linear or curvilinear if the amount of change in one variable does not bear a constant ratio to the amount of change in the other variable. For example if we double the amount of rainfall the production would not necessarily be doubled.

- (iii) **Simple, Partial and Multiple Correlation :** When only two variables are studied, it is called simple correlation.

If the common effect of two or more independent variables on one dependent data series is studied, it is called multiple correlations. For example if the study of rain, soil, temperature on potato production per acre is studied then it is multiple correlation.

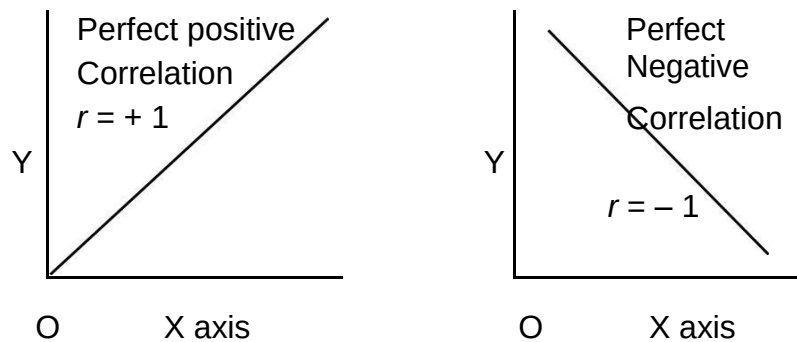
On the other hand, in partial correlation we recognize more than two variables, but consider only two variables to be influencing each other, the effect of other influencing variables being kept constant

Scatter Diagram:

It is the simplest method of studying the relationship between two variables diagrammatically. One variable is represented along the horizontal axis and the second variable along the vertical axis. For each pair of observations of two variables, we put a dot in the plane. There are as many dots in the plane as the number of paired observations of two variables. The direction of dots shows the scatter or concentration of various points. This will show the type of correlation.

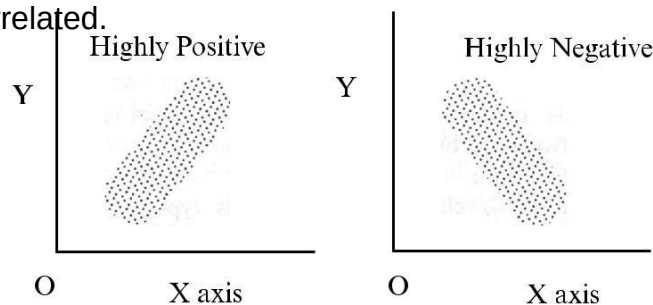
If all the plotted points form a straight line from lower left hand corner to the upper right hand corner then there is

Perfect positive correlation. We denote this as $r = +1$



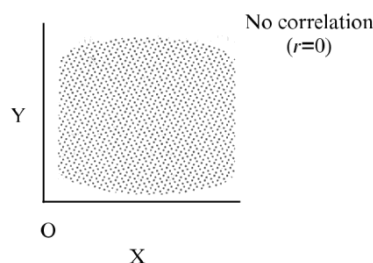
If all the plotted dots lie on a straight line falling from upper left hand corner to lower right hand corner, there is a perfect negative correlation between the two variables. In this case the coefficient of correlation takes the value $r = -1$.

If the plotted points in the plane form a band and they show a rising trend from the lower left hand corner to the upper right hand corner the two variables are highly positively correlated.



If the points fall in a narrow band from the upper left hand corner to the lower right hand corner, there will be a high degree of negative correlation.

If the plotted points in the plane are spread all over the diagram there is no correlation between the two variables.



Karl Pearson's Coefficient of Correlation (r) : Karl Pearson's Coefficient of Correlation is widely used in practice. It is an assumption of Karl Pearson's coefficient of correlation that linear relations exist in both the series.

A mathematical method for measuring the intensity or the magnitude of *linear relationship*

between two variables was suggested by Karl Pearson (1867-1936), a great British Biometrician and Statistician and, it is by far the most widely used method in practice.

Karl Pearson's measure, known as Pearsonian correlation coefficient between two variables X and Y , usually denoted by $r(X,Y)$ or r_{xy} or simply r is a numerical measure of linear relationship between them and is defined as the ratio of the covariance between X and Y , to the product of the standard deviations of X and Y .

This method is considered as the best measure because it provides the knowledge of directions of changes in data i.e. positive or negative, and also shows the degree of correlation which should always lie between +1 and -1.

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

$$= \frac{\text{Cov}(X, Y) \sum (X - \bar{X})(Y - \bar{Y})}{N}$$

$$\sigma_x = \frac{\sum (X - \bar{X})^2}{N}$$

$$\text{and } \sigma_y = \frac{\sum (Y - \bar{Y})^2}{N}$$

Direct Method	Short cut Method
Formula - (1) $r = \frac{\sum xy}{\sqrt{(\sum x^2 \times \sum y^2)}}$ (2) $r = \frac{\sum xy}{N \cdot \sigma_x \cdot \sigma_y}$ Here $x = X - \bar{X}$ & $y = Y - \bar{Y}$ N = No. of observations σ_x = Standard deviation of X-series σ_y = Standard deviation of Y-series	Formula - (1) $r = \frac{\sum dxdy - N(\bar{X} - A)(\bar{Y} - A)}{N \cdot \sigma_x \cdot \sigma_y}$ (2) $r = \frac{N \cdot \sum dxdy - (\sum dx \times \sum dy)}{\sqrt{[N \cdot \sum d^2x - (\sum dx)^2][N \cdot \sum d^2y - (\sum dy)^2]}}$ Here $dx = X - A_x$ & $dy = Y - A_y$ A = Assumed mean Or $r = \frac{N\sum xy - (\sum x)(\sum y)}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$ Where: N = number of pairs of scores $\sum xy$ = sum of the products of paired scores $\sum x$ = sum of x scores $\sum y$ = sum of y scores $\sum x^2$ = sum of squared x scores $\sum y^2$ = sum of squared y scores

SPEARMAN'S RANK CORRELATION

Sometimes we come across statistical series in which the variables under consideration are not capable of quantitative measurement but can be arranged in serial order. This happens when we are dealing with qualitative characteristics (attributes) such as honesty, beauty, character, morality, etc., which cannot be measured quantitatively but can be arranged serially. In such situations Karl Pearson's coefficient of correlation cannot be used as such. Charles Edward Spearman, a British Psychologist, developed a formula in 1904, which consists in obtaining the correlation coefficient between the ranks of N individuals in the two attributes under study.

Suppose we want to find if two characteristics A , say, intelligence and B , say, beauty are related or not. Both the characteristics are incapable of quantitative measurements but we can arrange a group of N individuals in order of merit (ranks) *w.r.t.* proficiency in the two characteristics. Let the random variables X and Y denote the ranks of the individuals in the characteristics A and B respectively. If we assume that there is no tie, *i.e.*, if no two individuals get the same rank in a characteristic then, obviously, X and Y assume numerical values ranging from 1 to N .

- (a) First assign the ranks to the data of both the series. Rank can be assigned by taking either the highest value as 1 or the lowest value as 1. But whether we start with the lowest value or the highest value we must follow the same method in case of both the variables.
- (b) Take the difference of two ranks ($R_1 - R_2$) *i.e.* (D)
- (c) Square the rank difference *i.e.* (D^2)

Spearman's rank correlation coefficient, usually denoted by ρ (Rho) is given by the equation

$$\rho = 1 - \frac{6 \sum d^2}{N(N^2 - 1)}$$

Where d is the difference between the pair of ranks of the same individual in the two characteristics and N is the number of pairs.

In case of Tied Ranks or Equal Ranks: If some values are equal in a distribution than average rank is given to those items. The formula for calculation of correlation will be :

$$R_P = 1 - \frac{6 \sum d^2 + \frac{1}{12} (m^3 - m) + \frac{1}{12} (m^3 - m) \dots \dots \dots}{N(N^2 - 1)}$$

M = No. of items where ranks are common.

Regression:

In business, several times it becomes necessary to have some forecast so that the management can take a decision regarding a product or a particular course of action. In order to make a forecast, one has to ascertain some relationship between two or more variables relevant to a particular situation. For example, a company is interested to know how far the demand for television sets will increase in the next five years, keeping in mind the growth of population in a certain town. Here, it clearly assumes that the increase in population will lead to an increased demand for television sets. Thus, to determine the nature and extent of relationship between these two variables becomes important for the company

WHAT IS REGRESSION

In 1889, Sir Francis Galton, a cousin of Charles Darwin published a paper on heredity, "*Natural Inheritance*". He reported his discovery that sizes of seeds of sweet pea plants appeared to "revert" or "regress", to the mean size in successive generations. He also reported results of a study of the relationship between heights of fathers and heights of their sons. A straight line was fit to the data pairs: *height of father versus height of son*. Here, too, he found a "regression to mediocrity" The heights of the sons represented a movement away from their fathers, towards the average height. We credit Sir Galton with the idea of statistical regression.

While most applications of regression analysis may have little to do with the "regression to the mean" discovered by Galton, the term "regression" remains. It now refers to the statistical technique of modeling the relationship between two or **more**

variables. In general sense, regression analysis means the estimation or prediction of the unknown value of one variable from the known value(s) of the other variable(s). It is one of the most important and widely used statistical techniques in almost all sciences - natural, social or physical.

INDEPENDENT AND DEPENDENT VARIABLES

Simple regression involves only two variables; one variable is predicted by another variable. *The variable to be predicted* is called the **dependent variable**. *The predictor* is called the **independent variable**, or *explanatory variable*. For example, when we are trying to predict the demand for television sets on the basis of population growth, we are using the demand for television sets as the dependent variable and the population growth as the independent or predictor variable.

The decision, as to which variable is which sometimes, causes problems. Often the choice is obvious, as in case of demand for television sets and population growth because it would make no sense to suggest that population growth could be dependent on TV demand! The population growth has to be the independent variable and the TV demand the dependent variable.

If we are unsure, here are some points that might be of use:

if we have control over one of the variables then that is the independent. For example, a manufacturer can decide how much to spend on advertising and expect his sales to be dependent upon how much he spends

if there is any lapse of time between the two variables being measured, then the latter must depend upon the former, it cannot be the other way round

if we want to predict the values of one variable from your knowledge of the other variable, the variable to be predicted must be dependent on the known one.

Why are there two Regression Lines? Under what conditions can there be only one Regression Line?

The task of bringing out linear relationship consists of developing methods of fitting a straight line, or a regression line as is often called, to the data on two variables.

The line of Regression is the graphical or relationship representation of the best estimate of one variable for any given value of the other variable. The nomenclature of the line depends on the independent and dependent variables. If X and Y are two variables of which relationship is to be indicated, a line that gives best estimate of Y for any value of X , it is called **Regression line of Y on X** . If the dependent variable changes to X , then best estimate of X by any value of Y is called **Regression line of X on Y** .

Regression Line X and Y : In this formation, Y is independent and X is

dependent variable, and best expected value of X is calculated corresponding to the given value of Y.

Regression Line Y on X : Here Y is dependent and X is independent variable, best expected value of Y is estimated equivalent to the given value of X.

Two Regression Line: An important reason of having two regression lines is that they are drawn on least square assumption which stipulates that the sum of squares of the deviations from different points to that line is minimum. The deviations from the points from the line of best fit can be measured in two ways – vertical, i.e. parallel to Y – axis, and horizontal i.e. parallel to X axis. For minimizing the total of the squares separately, it is essential to have two regression lines.

Single line of Regression : When there is perfect positive or perfect negative correlation between the two variables ($r = \pm 1$) the regression lines will coincide or overlap and will form a single regression line in that case.

Difference between Correlation & Regression :

Basics	Correlation	Regression
Relationship	Correlation tells the degree of average relationship between two or more variables.	Using relationship between known variables and unknown variables, the unknown variable is estimated
Cause & Effect	Unable to tell which series is the cause and which is the effect, despite high degree of correlation.	The given independent variable is the „cause“ and dependent variable is the effect.
Change in scale	Unaffected by the change of origin and scale.	Regression coefficient is affected by change in scale.
Application	Limited Application.	Wider Application.

Regression Equation X on Y	Regression Equation Y on X
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Original Form : $X = a + by$ $a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$ $b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$ Formula : $X - \bar{X} = by (Y - \bar{Y})$	Original Form : $Y = a + bx$ Formula : $Y - \bar{Y} = bx (X - \bar{X})$
Here - — X = Mean of x series. Y = Mean of y series. b_{xy} = Regression coefficient X on Y	Here - — X = Mean of x series. Y = Mean of y series. b_{yx} = Regression coefficient Y on X
Computation of Regression Coefficients :	
1. Direct Method : $b_{xy} = \frac{\sum xy}{\sum y^2}$ or $b_{xy} = r \frac{\sigma_y}{\sigma_x}$	1. Direct Method : $b_{yx} = \frac{\sum xy}{\sum x^2}$ or $b_{yx} = r \frac{\sigma_y}{\sigma_x}$
2. Short Cut Method : $b_{xy} = \frac{N \cdot \sum dxdy - (\sum dx \cdot \sum dy)}{N \cdot \sum d^2y - (\sum dy)^2}$ Here - dx = X - A dy = Y - A	2. Short Cut Method : $b_{xy} = \frac{N \cdot \sum dxdy - (\sum dx \cdot \sum dy)}{N \cdot \sum d^2x - (\sum dx)^2}$ Here - dx = X - A dy = Y - A

Business Forecasting:

Business forecasting is an act of predicting the future economic conditions on the basis of past and present information. It refers to the technique of taking a prospective view of things likely to shape the turn of things in foreseeable future. As future is always uncertain, there is a need of organised system of forecasting in a business.

BUSINESS FORECASTING is an estimate or prediction of future developments in business such as sales, expenditures, and profits. Given the wide swings in economic activity and the drastic effects these fluctuations can have on profit margins, it is not surprising that business forecasting has emerged as one of the most important aspects of corporate planning. Forecasting has become an invaluable tool for businesspeople to anticipate economic trends and prepare themselves either to benefit from or to counteract them. If, for instance, businesspeople envision an economic downturn, they can cut back on their inventories, production quotas, and hirings. If, on the contrary, an economic boom seems probable, those same businesspeople can take necessary measures to attain the maximum benefit from it. Good business forecasts can help business owners and managers adapt to a changing economy.

Thus, scientific business forecasting involves:

(i) Analysis of the past economic conditions and

(ii) Analysis of the present economic conditions; so as to predict the future course of events accurately.

In this regard, business forecasting refers to the analysis of the past and present economic conditions with the object of drawing inferences about the future business conditions. In the words of Allen, "Forecasting is a systematic attempt to probe the future by inference from known facts. The purpose is to provide management with information on which it can base planning decisions.

Leo Barnes observes, "**Business Forecasting is the calculation of reasonable probabilities about the future, based on the analysis of all the latest relevant information by tested and logically sound statistical econometric techniques, as interpreted, modified and applied in terms of an executive's personal judgment and social knowledge of his own business and his own industry or trade**".

Methods of Forecasting:

Quantitative Forecasting Techniques

Quantitative forecasts often use historical data, such as previous sales and revenue figures, production and financial reports and website traffic statistics. Looking at seasonal sales data, for example, can help a company plan next year's production and labor needs based on last year's monthly or quarterly figures. Quantitative forecasting also uses projections based on statistical modeling, trend analyses or other information from expert sources such as government agencies, trade associations and academic institutions.

1. Proforma Financial Statements:

Proforma statements use sales figures and costs from the previous two to three years after excluding certain one-time costs. This method is mainly used in mergers and acquisitions, as well as in cases where a new company is forming and statements are needed to request capital from investors.

2. Time-Series Forecasting:

Time-series forecasting is a popular quantitative forecasting technique, in which data is gathered over a period of time to identify trends. Time-series methods are one of the simplest methods to deploy and can be quite accurate, particularly over the short term. Some techniques that fall within this method are simple averaging and exponential smoothing.

3. Cause-Effect Method:

Here, the forecaster examines the cause-and-effect relationships of the variable with other relevant variables such as changes in consumers' disposable incomes, the interest rate, the level of consumer confidence, and unemployment levels. This method uses past time series on many relevant variables to produce the forecast for the variable of interest.

Qualitative Forecasting Techniques

Qualitative forecasting techniques come from the experience and instincts of seasoned business experts. These forecasting techniques aren't just guesses; they include interpretation of data combined with professional expertise developed during years on the job. For example, a company wanting qualitative information for projecting sales for the year might estimate the impact of a new ad campaign or promotion the company is planning, look at the effects that new technologies might have on consumer purchasing and take into account recent social fads and trends. A business might forecast demand by holding focus groups of customers to discuss and gauge their reactions to several new product features the company is considering.

1. Executive Opinions:

In this method, the expert opinions of key personnel of various departments, such as production, sales, purchasing and operations, are gathered to arrive at future predictions. The management team makes revisions in the resulting forecast, based on their expectations.

2. Reference Class Forecasting:

This method involves predicting the outcome of a planned action based on similar scenarios in other times or places. This is used to defy predictions that are arrived at based only on human judgment.

3. Delphi Technique:

Here, a series of questionnaires are prepared and answered by a group of experts, who are kept separate from each other. Once the results of the first questionnaire are compiled, a second questionnaire is prepared based on the results of the first. This second document is again presented to the experts, who are then asked to reevaluate their responses to the first questionnaire. This process continues until the researchers have a narrow shortlist of opinions.

4. Sales Force Polling:

Some companies believe that salespersons have close contact with the consumers and could provide significant insights regarding customer behavior. In this method of forecasting, the estimates are derived based on the average of sales force polling.

5. Consumer Surveys:

Businesses often conduct market surveys of consumers. The data is collected via telephonic conversations, personal interviews or survey questionnaires, and extensive statistical analysis is conducted to generate forecasts.

6. Scenario Writing:

In this method, the forecaster generates different outcomes based on diverse starting criteria. The management team decides on the most likely outcome from the numerous scenarios presented.

ANALYSIS OF TIME SERIES

When quantitative data are arranged in the order of their occurrence, the resulting statistical series is called a time series. The quantitative values are usually recorded over equal time interval daily, weekly, monthly, quarterly, half yearly, yearly, or any other time measure. Monthly statistics of Industrial Production in India, Annual birth-rate figures for the entire world, yield on ordinary shares, weekly wholesale price of rice, daily records of tea sales or census data are some of the examples of time series. Each has a common characteristic of recording magnitudes that vary with passage of time. Time series are influenced by a variety of forces. Some are continuously effective other make themselves felt at recurring time intervals, and still others are non-recurring or random in nature. Therefore, the first task is to break down the data and study each of these influences in isolation. This is known as decomposition of the

time series. It enables us to understand fully the nature of the forces at work. We can then analyse their combined interactions. Such a study is known as time-series analysis.

A time series is a sequential set of data points, measured typically over successive times. It is mathematically defined as a set of vectors $x(t), t = 0, 1, 2, \dots$ where t represents the time elapsed [21, 23, 31]. The variable $x(t)$ is treated as a random variable. The measurements taken during an event in a time series are arranged in a proper chronological order. A time series containing records of a single variable is termed as univariate. But if records of more than one variable are considered, it is termed as multivariate. A time series can be continuous or discrete. In a continuous time series observations are measured at every instance of time, whereas a discrete time series contains observations measured at discrete points of time. For example temperature readings, flow of a river, concentration of a chemical process etc. can be recorded as a continuous time series. On the other hand population of a particular city, production of a company, exchange rates between two different currencies may represent discrete time series. Usually in a discrete time series the consecutive observations are recorded at equally spaced time intervals such as hourly, daily, weekly, monthly or yearly time separations. As mentioned in [23], the variable being observed in a discrete time series is assumed to be measured as a continuous variable using the real number scale. Furthermore a continuous time series can be easily transformed to a discrete one by merging data together over a specified time interval.

Component of Time Series:

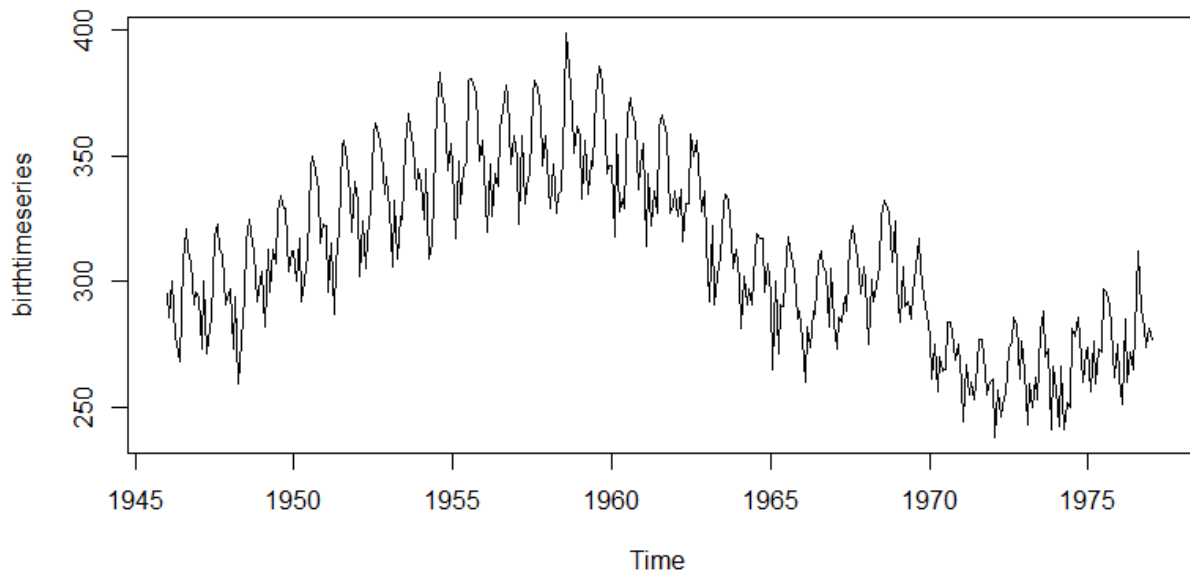
Traditional methods of time series analysis are concerned with *decomposing* of a series into a *trend*, a *seasonal variation* and other *irregular fluctuations*. Although this approach is not always the best but still useful (Kendall and Stuart, 1996).

The components, by which time series is composed of, are called component of time series data. There are four basic Component of time series data described below.

Different Sources of Variation are:

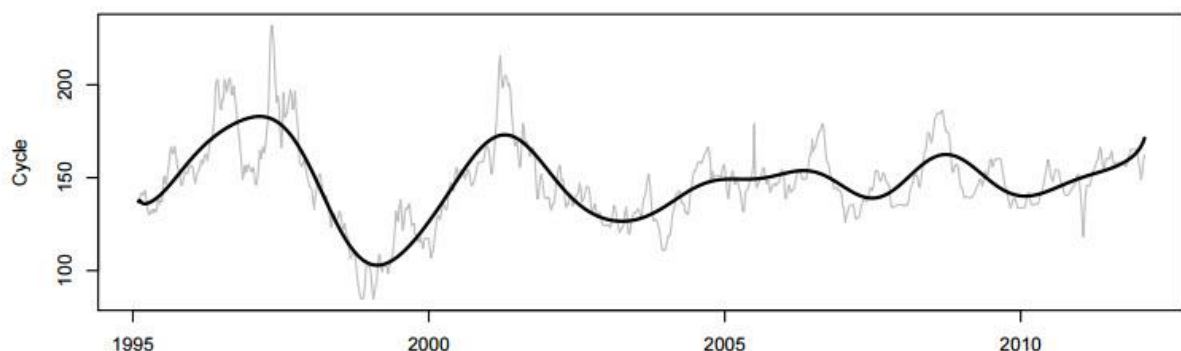
1. **Seasonal effect (Seasonal Variation or Seasonal Fluctuations)**
Many of the *time series data* exhibits a **seasonal variation** which is annual period, such as sales and temperature readings. This type of variation is easy to understand and can be easily measured or removed from the data to give *de-seasonalized data*. **Seasonal Fluctuations** describes any *regular variation (fluctuation)* with a period of less than one year for example cost of variation types of fruits and vegetables, cloths, unemployment figures, average daily rainfall, increase in sale of tea in winter, increase in sale of ice cream in summer etc., all show **seasonal variations**. The changes which repeat themselves within a fixed period, are also called **seasonal variations**, for example, traffic on roads in morning and evening hours, Sales at festivals like EID etc., increase in

the number of passengers at weekend etc. **Seasonal variations** are caused by climate, social customs, religious activities etc.



2. Other Cyclic Changes (Cyclical Variation or Cyclic Fluctuations)

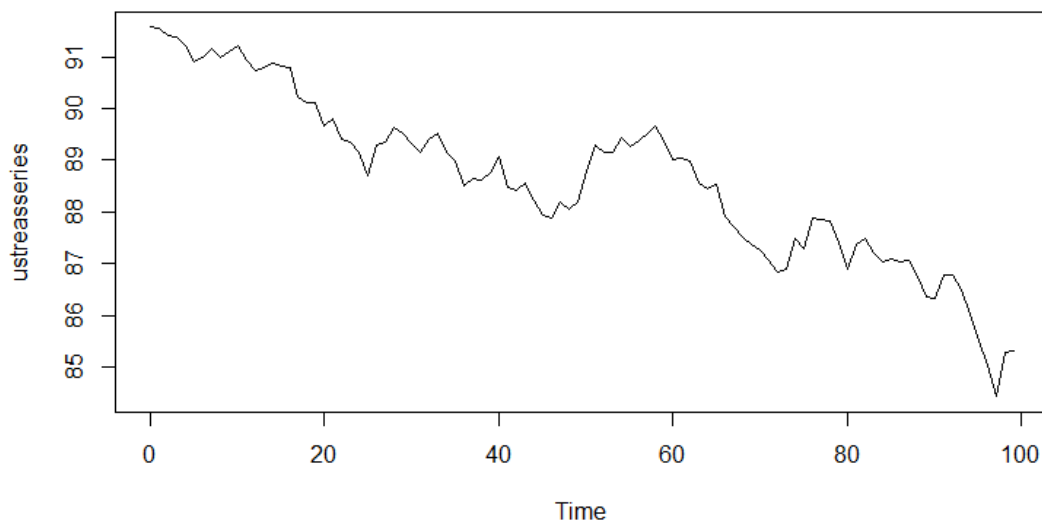
Time series exhibits **Cyclical Variations** at a fixed period due to some other physical cause, such as daily variation in temperature. **Cyclical variation** is a non-seasonal component which varies in recognizable cycle. sometime series exhibits oscillation which do not have a fixed period but are predictable to some extent. For example, economic data affected by business cycles with a period varying between about 5 and 7 years. In weekly or monthly data, the **cyclical component** may describes any *regular variation (fluctuations)* in time series data. The **cyclical variation** are periodic in nature and repeat themselves like business cycle, which has four phases (i) *Peak* (ii) *Recession* (iii) *Trough/Depression* (iv) *Expansion*.



3. Trend (Secular Trend or Long Term Variation)

It is a longer term change. Here we take into account the number of observations available and make a subjective assessment of what is **long term**. To understand the meaning of long term, let for example climate variables sometimes exhibit *cyclic variation* over a very long time period such as 50 years. If one just had 20 years data, this long term oscillation would appear to be a trend, but if

several hundreds years of data is available, then long term oscillations would be visible. These movements are systematic in nature where the movements are broad, steady, showing slow rise or fall in the same direction. The **trend** may be linear or non-linear (curvilinear). Some examples of secular trend are: Increase in prices, Increase in pollution, increase in the need of wheat, increase in literacy rate, decrease in deaths due to advances in science. Taking averages over a certain period is a simple way of detecting trend in seasonal data. Change in averages with time is evidence of a trend in the given series, though there are more formal tests for detecting **trend in time series**.



4. Other Irregular Variation (Irregular Fluctuations)

When *trend* and *cyclical variations* are removed from a set of time series data, the *residual* left, which may or may not be random. Various techniques for analyzing series of this type examine to see “if irregular variation may be explained in terms of probability models such as *moving average* or *autoregressive models*, i.e. we can see if any *cyclical variation* is still left in the **residuals**. These variation occur due to sudden causes are called **residual variation** (*irregular variation* or *accidental* or *erratic fluctuations*) and are unpredictable, for example rise in prices of steel due to strike in the factory, accident due to failure of break, flood, earth quick, war etc.

Decomposition of additive time series

